

Ghanithan Subramani

Principal Software Engineer

With over 13 years of experience spanning embedded systems, automotive software, and full-stack development, I specialize in designing and building high-performance, low-footprint software. My expertise lies in Rust, where I focus on pushing the boundaries of performance and reliability. I bring a T-shaped skill set, with deep knowledge in software architecture, coding (which I am passionate about), DevOps, testing, and UI/UX. I lead by inspiring collaboration within my team, fostering innovation and collective problem-solving.

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WORK EXPERIENCE

Principal Software Engineer

Astra Technologies Inc.

06/2023 - Present

- Software Architecture - Building high performance and low latency backend (REST and WS) using Rust
- Worked in setting up monitoring and traceability using Open telemetry, Jaeger, Prometheus, Grafana and Honeycomb.
- Frontend using Lit, Web Components, React, NextJS, Firebase- Building CI/CD pipelines using Github workflows

Senior Software Engineer

XRG Consulting Pvt Ltd - for Quickplay Media

09/2022 - 06/2023

- Building Golang based microservices as part of Content Management System for OTT platform provider Quickplay Media (Client: US media giant Allen Media Group [Localnow, The Weather Channel, Griot, etc]).
- Expertise includes EPG management, Image manipulation, VOD and Live Content Management Pipelines, Authorization and Entitlement Management.

Lead Fullstack Engineer

United India Insurance Company Ltd

01/2020 - 09/2022

- Built an internal Document Management System using GO (Gin) and mongoDB
- Managed Cross functional teams to deliver a holistic solution for Digilocker integration, claims reporting solution for banks and debugging critical issues.

Risk Underwriter

United India Insurance Company Ltd

11/2016 - 12/2019

- Was inducted as a Fellow of Insurance Institute of India with distinction in Actuary (Reg: FE93778)

Senior Software Engineer

Bosch Global Software Technologies

09/2014 - 09/2016

- Implemented Injection over CAN for Diesel Truck ECU.
- Package responsible for Hardware Encapsulation layer

Sr. Design and Development Engineer

HandsOn Technologies

04/2013- 08/2014

- Created libraries for communication protocols like CAN, I2C , SPI , UART.
- Written firmware for products based on ARM Cortex-M0.
- Was involved in complete product development cycle from requirement analysis till testing of the product.
- Designed 10+ PCBs during my tenure which are still in use today.

Software Engineer

Bosch Global Software Technologies

06/2011- 03/2013

- Implemented device drivers for sensors.
- Contributed in resource optimization and involved in solving two major bugs in the product.

Technical Skills

Languages

Rust, GO, Javascript, C, Python, Java, C++

DBMS

Postgres, Couchbase, Oracle, MySQL, mongoDB

Web Technologies

API (WS, REST), JWT, PWA, Bitcoin Wallet, CMS - Streaming, EPG Management, Image Processing, SPA, Otel, Kubernetes

Frameworks

Tokio, Axum, Actix, NextJS, Lit, Svelte, ExpressJS, Fiber

Tools

Figma, Prometheus, Jaeger, Grafana, vite, git, Swagger API, PlantUML, Postman, Docker

Cloud Platforms

AWS (complete VPC setup) , GCP

Education

Bachelor's in Electronics and communication

Kumaraguru College of Technology

06/2007 - 04/2011

Projects

- Working on URLJammer, a url shortener based on Jamstack
- Interactive console mp3player written in C++
- PDFReplicator used to create millions of PDFs from a template and a data source, built in JAVA
- Intelligent Traffic Control System using Zigbee Mesh
- Integrated Brake and Turn Indication System (1st prize in HW section in VISION '10 - Anna University)